Claudio Fanconi

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Education

University of Cambridge *Cambridge, United Kingdom*

Ph.D. Artificial Intelligence and Machine Learning *4/2024 - 04/2028 (expected)*

- Supervisor: Mihaela van der Schaar
- Researching LLM reasoning, (inverse) reinforcement learning, and alignment for decision-making
- Publications: 2 NeurIPS first-author, 3 ICML/ICLR workshop first-author (2 oral presentations)
- Published paper topics: (I)RL for LLM reasoning, LLM model efficiency, LLM preference learning

ETH Zürich *Zürich, Switzerland*

M.Sc. Information Technology & Electrical Engineering *9/2020 - 10/2022*

- Focus: Machine learning and signal processing with Master's thesis at **Stanford University** (3/2022 - 10/2022)
- Publications: 1 EMNLP first-author, 1 first-author journal, 1 ICML workshop first-author
- Published paper topics: Deep learning interpretability, Bayesian uncertainty estimation, NLP for healthcare
- GPA: 5.53/6.00

ETH Zürich *Zürich, Switzerland*

B.Sc. Information Technology & Electrical Engineering *9/2016 - 8/2019*

- Studied abroad at the **Chinese University of Hong Kong** during the autumn semester of 2018
- Relevant subjects: data structures & algorithms, probability & statistics, calculus, linear algebra, machine learning

Professional Experience

Google DeepMind *Zürich, Switzerland*

Research Scientist Intern *4/2026 - 10/2026*

- Researched multi-modal reasoning and latent thoughts for Gemma models for visual tasks
- Authored one paper and co-invented three patents on visual reasoning that are currently in the filing process

Sony AI *Zürich, Switzerland*

Research Scientist Intern *3/2023 - 12/2023*

- Worked on the "ACE" project, developing an autonomous superhuman table tennis robot (Nature Article)
- Researched deep learning models for object detection and velocity prediction, achieving < 2px error
- Deployed proposed model in a system using TensorRT in C++ to achieve real-time predictions with sub-2ms latency

McKinsey & Company *Zürich, Switzerland*

Management Consulting Intern *6/2020 - 8/2020*

- Implemented an algorithmic pricing engine for a retail company in Python to price 60K products quarterly
- Developed impact scenarios of sales revenues, presented and explained these findings to non-technical stakeholders

IBM *Böblingen, Germany*

Machine Learning Engineer Intern *10/2019 - 03/2020*

- Prototyped near-duplicate text detection using TensorFlow neural networks, reducing matching time by 50%
- Co-invented five patents on near-duplicate detection and data quality (Google Patents)

Skills

Programming: Python (proficient), Java (advanced), C++ (intermediate), SQL (intermediate)

Languages: German (native), Italian (native), English (proficient), Romansh (advanced), French (basic)

Conference Publications

C. Fanconi, M. van der Schaar - 2025
Cascaded Language Models for Cost-effective Human-AI Decision-Making
NeurIPS 2025 (& ICML 2025 Workshop on Multi-Agent Systems) (Paper, GitHub, YouTube)

C. Lu*, S. Holt*, **C. Fanconi***, A.J. Chan, J. Foerster‡, M. van der Schaar‡, R. T. Lange‡ - 2024
Discovering Preference Optimization Algorithms with and for Large Language Models
NeurIPS 2024 (& ICML 2024 Workshop on AutoRL) (Paper, GitHub, Blog) *Equal contribution

C. Fanconi*, M. Vandenhirtz*, S. Husmann, and J. Vogt - 2023
This Reads Like That: Deep Learning for Interpretable Natural Language Processing
EMNLP 2023. (Paper, GitHub, YouTube) *Equal contribution

C. Fanconi, M. van Buchem, and T. Hernandez-Boussard - 2023
Natural Language Processing Methods to Identify Oncology Patients at High Risk for Acute Care with Clinical Notes
AMIA 2023 Informatics Summit (Paper, GitHub)

Workshop Publications

C. Fanconi, N. Astorga, M. van der Schaar - 2026
Learning Reasoning Rewards from Expert Demonstration via Inverse Reinforcement Learning **(Outstanding paper award)**
ICLR 2026 Workshop on LLM Reasoning & **ICML 2026** Workshop on Decision-Making from Offline Datasets
(Paper, GitHub)

P. Rauba, **C. Fanconi**, M. van der Schaar - 2026
Tiny Autoregressive Recursive Models
ICLR 2026 Workshop on Latent & Implicit Thinking and Workshop on Recursive Self-Improvement **(Spotlight)**

K. Kobalczyk*, **C. Fanconi***, H. Sun, M. van der Schaar - 2025
Few-shot Steerable Alignment: Adapting Rewards and LLM Policies with Neural Processes
ICML 2025 Workshop on Human Feedback for AI Alignment **(Oral, top 10%)** (Paper, GitHub) *Equal contribution

A. Hoffmann*, **C. Fanconi***, R. Rade*, J. Kohler - 2021
This looks like that... Does it? Shortcomings of Latent Space Prototype Interpretability in Deep Networks
ICML 2021 Workshop on Explainable AI (Paper, GitHub, YouTube) *Equal contribution

Journal Publications

M. van Buchem, A. de Hond, **C. Fanconi**, V. Shah, M. Schüszler, I. Kant, E. Steyerberg, and T. Hernandez-Boussard - 2024
Applying Natural Language Processing to Patient Messages to Identify Depression Concerns in Cancer Patients
Journal of the American Medical Informatics Association (Paper)

A. de Hond, M. van Buchem, **C. Fanconi**, M. Roy, D. Blayney, I. Kant, E. Steyerberg, and T. Hernandez-Boussard - 2024
Predicting Depression Risk in Patients With Cancer Using Multimodal Data: Algorithm Development Study
JMIR Medical Informatics (Paper, GitLab)

C. Fanconi, A. de Hond, D. Peterson, A. Capodici, and T. Hernandez-Boussard - 2023
A Bayesian Approach to Predictive Uncertainty in Chemotherapy Patients at Risk of Acute Care
The Lancet eBioMedicine (Paper, GitHub)

F. Sanmarchi, **C. Fanconi**, D. Golinelli, D. Gori, T. Hernandez-Boussard, and A. Capodici - 2023
Predict, Diagnose, and Treat Chronic Kidney Disease with Machine Learning: a Systematic Literature Review
Journal of Nephrology (Paper)